Roll Number: _____ Name: _____

Thapar Institute of Engineering & Technology, Patiala

Department of Computer Science and Engineering

END Sem Examination

Max. Marks: 40

Time: 03 Hours

Date: 23 July 2025, 2:00 PM to 5:00 PM

Course Name: Database Management System

Course Code: UCS310

B.E- Summer Sem

Name of Faculty: Dr. Vibha Jain

Note: ***Attempt any four questions.***

Assume missing data, if any, suitably and mention the same. Ignore extra spaces, typing mistakes etc.

All the answer should be attempted sequentially.

Q.No	Questions	Marks
Q1	a) Differentiate the following with suitable example: i) Delete vs truncate commands in SQL ii) Correlated sub-query vs non-correlated sub-query iii) On delete cascade vs On delete set null iv) Weak entity vs strong entity v) Physical data independence vs logical data independence	2*5
Q2	a) A library system maintains information about books, members, and staff. Each book has a unique ID, title, and author. Members can borrow multiple books. Staff members issue and return books for members. • Identify entities, attributes, and relationships. • Convert the E-R diagram into relational tables. b) Identify which SQL clause belongs to DML, DDL, and DQL. Create a table with three columns: DDL, DML, and DQL, and categorize the given SQL commands accordingly. DROP, RENAME, UPDATE, DELETE, SELECT, TRUNCATE, ALTER, INSERT c) Specify the type of mapping cardinalities for the following example scenarios: i) Each employee is assigned to exactly one department, but a department can have multiple employees working in it ii) A student can enroll in multiple courses, and a course can have multiple students enrolled in it.	4 4 2
Q3	a) Discuss the differences and similarities between third normal form and Boyce coded Normal form with suitable examples. b) Consider a relation R (A, B, C). Find the canonical cover of the following functional dependencies holds in R: A -> BC, B -> C, A -> B, AB -> C. c) Consider the relation R(ABCDE) with following dependencies A->B, BC->E, and ED->A i) List all the candidate keys of R. ii) Identify the normal form of R, by showing all the intermediate steps involved (check up to 3NF).	3 3 4
Q4	a) List and briefly explain the ACID properties of a transaction in a database system. Also, draw the state diagram of a transaction, describing each state and the transitions involved. b) Define recoverable schedule and cascade-less schedule. Differentiate between them with suitable examples.	4 2

	c) What is the difference between conflict serializability and view serializability? Check whether the schedule S is conflict serializable. Justify your answer by constructing a precedence graph. S : R1 (A) , R2(A) , R1 (B) , R2(B) , R3(B) , W1 (A) , W2(B).	4
Q5	a) What is the Two-Phase Locking (2PL) protocol? Explain with the help of a suitable example how 2PL ensures conflict-serializability, but a schedule following 2PL may still not be cascade-less. b) Write short notes on i. Log based recovery mechanism ii. Checkpoint iii. Strict 2PL vs Rigorous 2PL	4 6

********** **All the best** **********